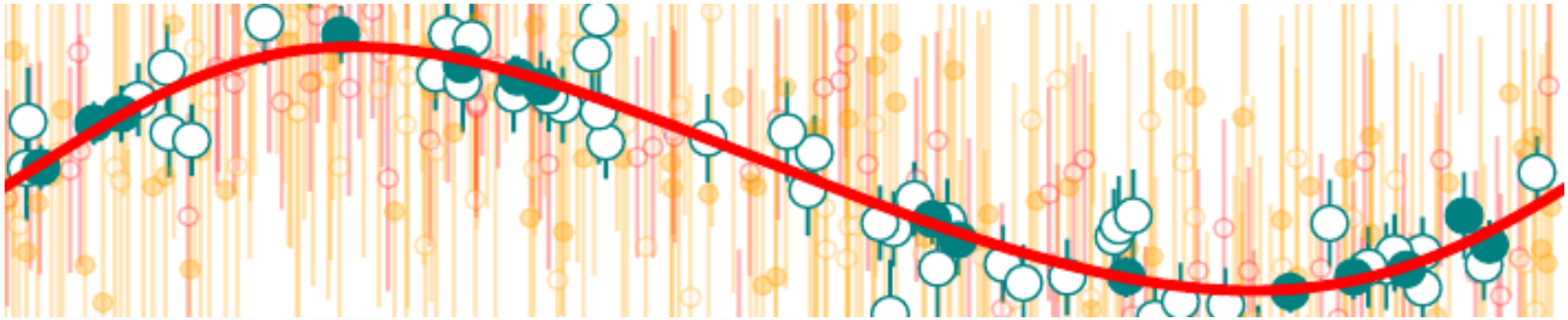


Fitting astronomical data to a model

M1 CCP HAP 703P

Atelier Astrophysique 1

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04 novembre 2026

FACULTÉ DES SCIENCES DE MONTPELLIER

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Outline

1 Model fitting in astronomy

2 From least squares to Levenberg-Marquardt

3 Other topics in statistics and modeling

4 Bibliographic ressources

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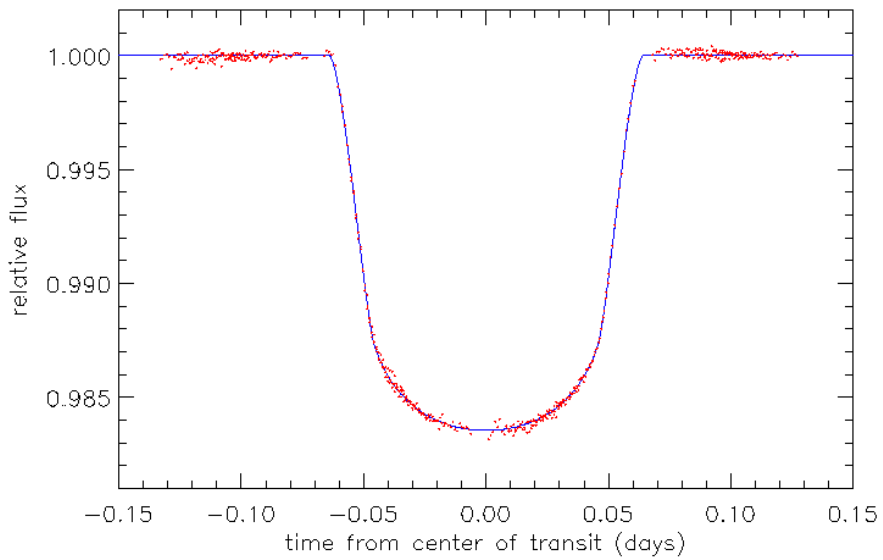
The data modeling problem

- Knowing...
  - N measurements $\{y_i\}$
  - taken at abscissa $\{x_i\}$
  - w/ uncertainties $\{\sigma_i\}$
- Assuming...
  - model function f depending on M parameters $\{a_j\}$

➔ What is the most-likely set of parameters $\{a_j\}$ that matches our measurements $\{x_i, y_i\}$ w/ uncertainties $\{\sigma_i\}$?

➔ What are the uncertainties on $\{a_j\}$?

➔ Is the fit statistically good?



Fit of photometric data: transit lightcurve of HD 209458b

Brown et al. (2001)

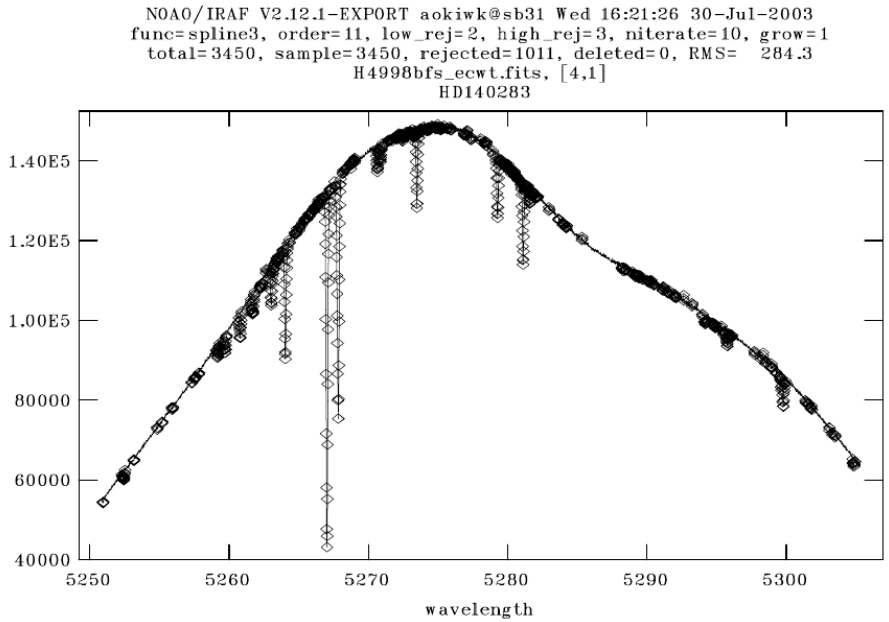
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Fit of a spectrum continuum

Aoki (2008)

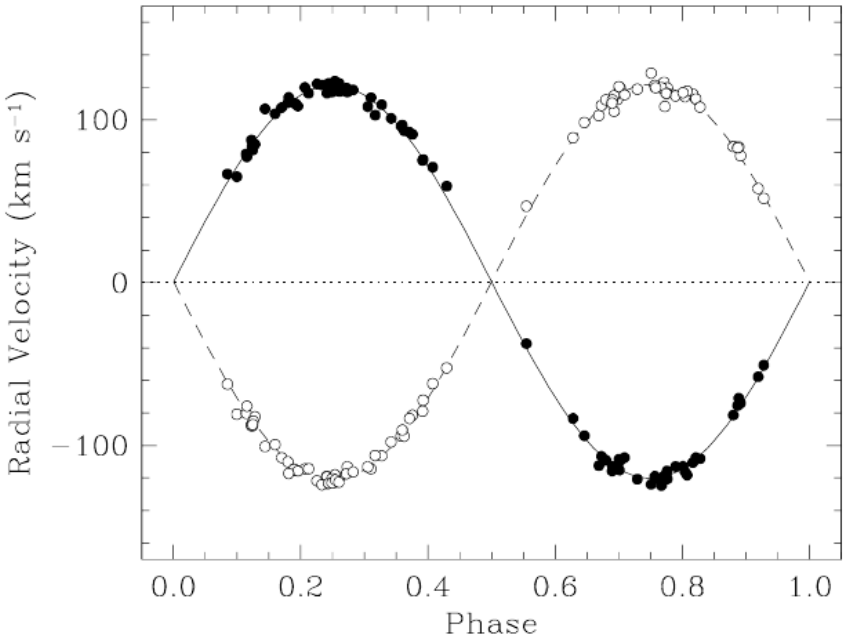
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Fit of spectroscopic data: radial velocities of the binary system YY Gem

Torres & Ribas (2002)

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Element	Value
Adjusted quantities:	
P (days)[a]	0.814282212
γ (km s^{-1})	0.54 ± 0.26
K_A (km s^{-1})	121.18 ± 0.42
K_B (km s^{-1})	120.51 ± 0.42
e[b]	0
Min I (HJD $-2{,}400{,}000$)[a]	49,345.112327
Derived quantities:	
$M_A \sin^3 i\ (M_\odot)$	0.5938 ± 0.0046
$M_B \sin^3 i\ (M_\odot)$	0.5971 ± 0.0046
$q \equiv M_B/M_A$	1.0056 ± 0.0050
$a_A \sin i\ (\times 10^6$ km)	1.3569 ± 0.0047
$a_B \sin i\ (\times 10^6$ km)	1.3493 ± 0.0047
$a \sin i\ (R_\odot)$	3.8882 ± 0.0095
Other quantities pertaining to the fit:	
N_{obs}	90
Time span (days)	456.9
σ_A (km s^{-1})	3.44
σ_B (km s^{-1})	3.44

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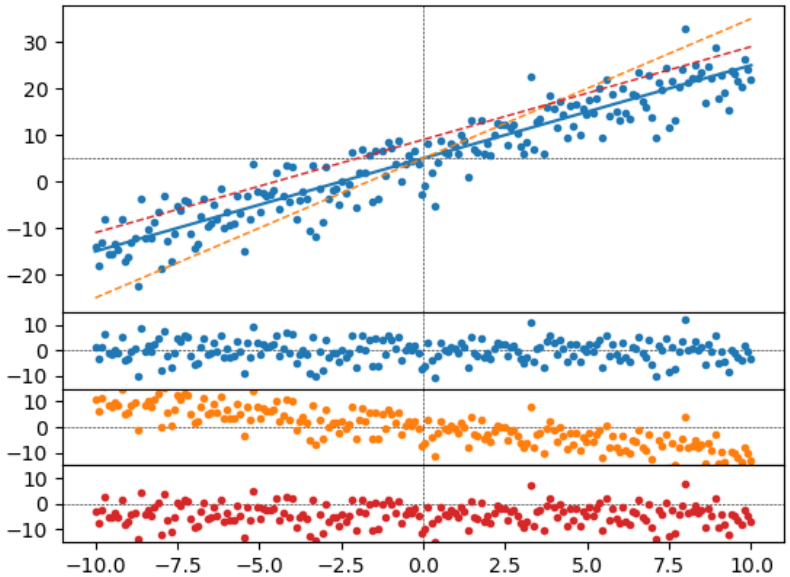
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Measurement of goodness of fit

- Goodness-of-fit ➔ residuals
  - Mean square error
    - $MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - f(x_i))^2$
  - Root mean square
    - $RMS = \sqrt{MSE}$
- "Good model"
  - Residuals compatible w/ noise
  - $RMS = \sigma$



$RMS = 4.13$

$RMS = 7.31$

$RMS = 5.83$

Minimum least squares as maximum likelihood

- We define the χ^2 as a measure of the goodness-of-fit or "merit function"

$$\chi^2 = \sum_{i=1}^{N} \left(\frac{y_i - f(x_i)}{\sigma_i} \right)^2$$

- We identify...
  - the probability of measuring the data $\{y_i\}$ given the set of parameters $\{a_j\}$
    - what we can compute
  - as the likelihood of the parameters given the data
    - what we seek to determine
- Maximum likelihood
  - Given our model, assuming that y_i is a Gaussian random variable with mean $\mu = f(x_i)$ and standard deviation σ_i:

$$P(y_i|\{a_j\}) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[\frac{-(y_i - f(x_i))^2}{2\sigma_i^2} \right]$$

  - For independant measurements:

$$L = P(\{y_i\}|\{a_j\}) = \prod_{i=1}^{N} P(y_i|\{a_j\}) \quad \Rightarrow \quad \ln(L) = \ln \alpha - \ln \chi^2$$

Statistical meaning of a χ^2 value

- We obtain a fit with a final (minimal) value of χ^2, what does it mean?
  - For models linear in $\{a_j\}$ and normally distributed errors, the probability distribution for different values of χ^2 at its minimum is the chi-squared distribution for $\nu = N - M$ **degrees of freedom**
  - Probability that the observed chi-squared exceeds a measured value χ^2 by chance is : $Q(\chi^2|\nu) = Q(\frac{\nu}{2}, \frac{\chi^2}{2}) = 1 - P(\frac{\nu}{2}, \frac{\chi^2}{2})$
    - with P the incomplete Γ function and $Q = \dfrac{\Gamma(a,x)}{\Gamma(a)}$
  - For a Statistically good fit:

$$\chi^2 \simeq \nu \text{ or : } \chi^2_\nu \simeq 1 \text{ with: } \chi^2_\nu = \frac{\chi^2}{\nu}$$

  - with standard deviation:

$$\sqrt{2\nu} \text{ on } \chi^2 \text{ or } \sqrt{\frac{2}{\nu}} \text{ on } \chi^2_\nu \text{ for } \nu \gg 1$$

- Also applied to non-linear models
- Requires reliable estimates of $\{\sigma_i\}$ and absence of outliers

χ^2 minimisation methods

- Model linear in $\{a_j\}$ (not necessarily in x!)
  - polynomial, Fourier series decomposition
  - derivatives $\frac{\partial \chi^2}{\partial a_j}$ are constant
  - matricial equation solved w/ LU decomposition, Gauss-Jordan elimination...
- Non-linear model
  - solution found iteratively (steepest gradient method)
  - Levenberg-Marquardt algorithm: uses 1st and 2nd derivatives of χ^2 vs $\{a_j\}$
  - stop criterion: $\Delta\chi^2 \ll 1$
  - issue of local minima ➔ importance of initial guess
- Outputs
  - estimated parameters $\{\hat{a}_k\}$
  - final merit function value χ^2_f
  - variance-covariance matrix C_{jk}
    - uncertainties on parameters: $\sigma(a_j) = \sqrt{C_{jj}}$
    - if error bars require rescaling: $\sigma(a_j) = \sqrt{\chi^2_\nu}\sqrt{C_{jj}}$

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Other topics in statistics and modeling

- Statistical tests
  - Spearman Rank Correlation, Students-t test, Kolmogorov-Smirnov (KS) test
  - "Modern statistical methods for astronomy", E.D. Figelson & G.J. Babu
  - "Statistics done wrong"
- Period search
  - "Understanding the Lomb-Scargle Periodogram", J.T. VanderPlas [arXiv:1703.09824]
- Inverse problems and regularisation
  - See P. Lena et al. in the bibliography

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- *"Numerical Recipes in C – The Art of Scientific Programming"*, W.H. Press et al., 1992, Cambridge University Press
  ISBN-13: 0-521-43108-5
  - Chapter 15: "Modeling of data"
  - §6.2: "Incomplete Gamma Function, Error Function, Chi-Square probability Function, Cumulative Poisson function"
- *"Data analysis recipes: Fitting a model to data"*, D.W. Hogg, `[arXiv:1008.4686]`
- *"L'Observation en Astrophysique"*, P. Léna et al., 2008, EDP Sciences
  ISBN-13: 978-2868838773
  - §9.5 : "Le problème de l'estimation"